Chapter 5

PRINCIPLES OF COUNTING

EXERCISE 5

Short Questions

Q.2 Solve / write answers of the following short questions:

(i) How many different four-letter initials can people have?

Sol: Four-letter initials are determined by specifying the first letter (26 ways), then the second letter (26 ways), then the third letter (26 ways), and then the fourth letter (26 ways).
Therefore by the product rule, possible four-letter initials are

$$26 \cdot 26 \cdot 26 \cdot 26 = 26^4 = 456{,}976$$

(ii) How many different four-letter initials with none of the letters repeated can people have?

Sol: Since none of the letters repeated, so four-letter initials are determined by specifying the first letter (26 ways), then the second letter (25 ways), then the third letter (24 ways), and then the fourth letter (23 ways).
Therefore by the product rule, possible four-letter initials are

$$26 \cdot 25 \cdot 24 \cdot 23 = 358{,}800$$

(iii) How many different four-letter initials are there that begin with an *A*?

Sol: There is only one way to specify the first initial, but, there are 26 ways to specify each of the other initials. Therefore there are, by the product rule, possible four-letter initials beginning with *A* are

$$1 \cdot 26 \cdot 26 \cdot 26 = 26^3 = 17{,}576$$

(iv) How many different four-letter initials are there that end with *Z*?

Sol: There is only 1 way to specify the last initial, but, there are 26 ways to specify each of the other initials. Therefore there are, by the product rule, possible four-letter initials ending with *Z* are

month?

Sol: Assuming no were born on 29th February (leap day), we have

$N = 47$

$k = 12$

Using generalized pigeonhole principle, we have at least

$$\left\lceil \frac{N}{k} \right\rceil = \left\lceil \frac{47}{12} \right\rceil = \lceil 3.92 \rceil = 4$$

people who were born in the same month.

(xiii) Write $\frac{5\cdot 3}{8\cdot 6\cdot 7}$ in factorial form.

PU. 2016 (BS Math)

Sol: $\frac{5\cdot 3}{8\cdot 6\cdot 7} = \frac{15}{8\cdot 7\cdot 6} = \frac{15(5!)}{8\cdot 7\cdot 6\cdot 5!} = \frac{15(5!)}{8!}$

(xiv) Find the value of each of these quantities.

(a) $P(6, 3)$ **(b)** $P(6, 5)$ **(c)** $P(8, 1)$
(d) $P(8, 5)$ **(e)** $P(8, 8)$ **(f)** $P(10, 9)$

Sol:(a) $P(6,3) = \frac{6!}{(6-3)!} = \frac{6!}{3!} = \frac{720}{6} = 120$

(b) $P(6,5) = \frac{6!}{(6-5)!} = \frac{6!}{1!} = \frac{720}{1} = 720$

(c) $P(8,1) = \frac{8!}{(8-1)!} = \frac{8!}{7!} = 8$

(d) $P(8,5) = \frac{8!}{(8-5)!} = \frac{8!}{3!} = \frac{40320}{6} = 6720$

(e) $P(8,8) = 8! = 40320$

(f) $P(10,9) = \frac{10!}{(10-9)!} = \frac{10!}{1!} = \frac{3628800}{1} = 3628800$

(xv) Find the value of each of these quantities.

(a) $C(5, 1)$ **(b)** $C(5, 3)$ **(c)** $C(8, 4)$
(d) $C(8, 8)$ **(e)** $C(8, 0)$ **(f)** $C(12, 6)$

Sol:(a) $C(5,1) = \frac{5!}{(5-1)!1!} = \frac{5!}{4!} = 5$ **(b)** $C(5,3) = \frac{5!}{(5-3)!3!} = \frac{5!}{2!3!} = 10$

(c) $C(8,4) = \frac{8!}{(8-4)!4!} = \frac{8!}{4!4!} = 70$

(d) $C(8,8) = \frac{8!}{(8-8)!8!} = \frac{8!}{0!8!} = 1$ **(e)** $C(8,0) = \frac{8!}{(8-0)!0!} = \frac{8!}{8!} = 1$

(f) $C(12,6) = \frac{12!}{(12-6)!6!} = \frac{12!}{6!6!} = 924$

(xvi) State Binomial theorem.

PU. 2017, 2016, 2015, 2013 (BS Math)

Sol: **Binomial Theorem:**

Statement: Let x and y be variables, and let n be a nonnegative integer, then

$$(x+y)^n = \sum_{j=0}^{n}\binom{n}{j}x^{n-j}y^j = \binom{n}{0}x^n + \binom{n}{1}x^{n-1}y + \ldots + \binom{n}{n-1}xy^{n-1} + \binom{n}{n}y^n$$

(xvii) Find the sum of all two digit positive integers which are neither divisible by 5 nor by 2. PU, 2017 (BS Math)

Sol: Two digit positive integers which are neither divisible by 5 nor by 2 are

11, 13, 17, 19, 21, 23, 27, 29, 31, 33, 37, 39, ..., 91, 93, 97, 99

Sum of these numbers can be written as

(11 + 13 + 17 + 19) + (21 + 23 + 27 + 29) + (31 + 33 + 37 + 39) + ... + (91 + 93 + 97 + 99)

or 60 + 100 + 140 + ... + 380

This is an arithmetic series with

$a_1 = 60,\ d = 100 - 60 = 40,\ a_n = 380$

$$a_n = a_1 + (n-1)d \qquad\qquad 320 = 40(n-1)$$
$$380 = 60 + (n-1)(40) \qquad\qquad n - 1 = 8$$
$$380 - 60 = 40(n-1) \qquad\qquad n = 9$$

Hence sum of such numbers is given by

$$S_n = \frac{n}{2}[2a_1 + (n-1)d]$$

$$S_9 = \frac{9}{2}[2(60) + (9-1)(40)] = \frac{9}{2}[120 + 320] = \frac{9}{2}[440] = 1980$$

(xviii) Prove that $|x+y| \le |x| + |y|;\ \forall x, y \in R$. PU, 2015, 2012 (BS Math)

Sol: Since $|x|^2 = x^2\ \ \forall x \in R$, so

$$|x+y|^2 = (x+y)^2 = x^2 + 2xy + y^2$$
$$\le x^2 + 2|xy| + y^2 \qquad \because xy \le |xy|$$
$$= x^2 + 2|x||y| + y^2 \qquad \because |xy| = |x||y|$$
$$= |x|^2 + 2|x||y| + |y|^2$$
$$= [|x| + |y|]^2 \qquad \because |x|^2 = x^2, |y|^2 = y^2$$

$$\Rightarrow |x+y|^2 - [|x| + |y|]^2 \le 0$$
$$\Rightarrow [|x+y| - [|x| + |y|]][|x+y| + [|x| + |y|]] \le 0$$
$$\Rightarrow |x+y| - [|x| + |y|] \le 0$$
$$\Rightarrow |x+y| \le |x| + |y| \quad \forall x, y \in R$$

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